



Department for
Science, Innovation
& Technology

Aligning Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) setup guidance in AWS with Secure by Design

February 2026

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Background

In AWS environments, you can implement Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) capabilities using a combination of native services and partner solutions. Integrating these services with centralised logging, alerting and automation allows government organisations to:

- detect anomalous activity early
- enforce preventative controls at scale
- maintain the agility and shared responsibility model inherent to cloud architectures

The [Secure by Design](#) approach is a government framework which embeds cyber security into digital services from the outset. Its goal is to make security an integral, proactive part of all government digital systems, rather than an afterthought.

Purpose

The purpose of this document is to align IDS and IPS deployment guidance with the government's Secure by Design approach.

For systems hosted on Amazon Web Services (AWS), this guidance supports the use of cloud-native and integrated IDS and IPS capabilities as part of a defence-in-depth approach. You can use AWS services such as [Amazon GuardDuty](#), [AWS Network Firewall](#) and [AWS Web Application Firewall \(WAF\)](#) alongside appropriate configuration, monitoring and response processes to support Secure by Design principles.

The way you integrate IDS and IPS controls should be consistent, repeatable and reliable. This approach strengthens defensive capability, improves resilience to cyber threats and makes sure controls are implemented in an effective, scalable manner that's aligned with wider organisational security objectives.

Secure by Design on the UK Government Security website

- [Secure by Design principles](#)
- [Secure by Design policy](#)
- [Implementing Secure by Design](#)
- [Guide to adopting Secure by Design](#)

1. IDS and IPS in AWS environments

Since AWS networking and workloads differ fundamentally from on-premises environments, IDS and IPS must use cloud-native features and architecture. This includes [AWS CloudTrail](#), and Amazon Virtual Private Cloud (VPC) features like [VPC Flow Logs](#) and [Traffic Mirroring](#).

Detection typically uses passive telemetry to identify anomalies without affecting a system's availability. You should use prevention, which enforces controls inline, proportionally, based on risk. This reduces the attack surface while maintaining operational flexibility.

This approach supports the following Secure by Design principles:

- 4. Design usable security controls
- 7. Minimise the attack surface

IDS and IPS should also be integrated with centralised logging and monitoring solutions to give a unified view of threats across multiple AWS accounts or regions. If you use services like [AWS Security Hub Cloud Security Posture Management \(CSPM\)](#) or [Amazon EventBridge](#) you can correlate and prioritise alerts.

It's also worth considering workload-specific IDS or IPS deployment, because different workloads, such as EC2 instances, containers or serverless services, may require different detection approaches. The controls should be tailored to their operational context so they're easy to use and efficient.

This approach supports the following Secure by Design principles:

- 3. Adopt a risk-driven approach
- 4. Design flexible architectures
- 5. Build in detect and respond security

2. IDS and IPS governance

For the Secure by Design approach to be successful, cyber security risks must be clearly assigned, understood and actively managed.

Within AWS environments, you should clearly define responsibility for IDS and IPS capabilities across architectural design, operational management and incident response functions. Senior owners should have overall accountability, while operational teams will be responsible for deploying, operating, and monitoring detection and prevention controls.

Secure by Design offers an [example RACI matrix](#) showing roles and responsibilities, but responsibility and accountability is likely to vary based on each organisation's setup and governance structure.

AWS recommends using AWS Security Hub CSPM to aggregate security findings and track remediation activities. This provides continuous assurance and ongoing risk management.

Formal governance structures should establish clear escalation routes, decision-making authority and routine review cycles so that controls remain effective over time. And by embedding secure design considerations into architectural review boards and changing advisory processes, you can reinforce long-term operational resilience even more.

This approach supports the following Secure by Design principles:

- 1. Create responsibility for cyber security risk
- 5. Build in detect and respond security
- 9. Embed continuous assurance

3. AWS-native IDS and IPS capabilities

AWS offers managed detection and prevention services that reduce operational complexity and embed secure-by-default controls. Amazon GuardDuty continuously monitors accounts, workloads and network activity to detect threats. This supports Secure by Design principles that require detection to be built-in, maintained and updated automatically.

Preventative services such as AWS Network Firewall and AWS WAF enforce controls at the network and application layers. These managed services:

- reduce supply chain risk
- provide layered defence
- ensure enforcement aligns with detection insights

AWS-native services also benefit from automated threat intelligence updates. GuardDuty and WAF automatically incorporate global threat feeds and anomaly detection, so that IDS and IPS controls respond to emerging threats without manual intervention.

Native AWS tools also simplify audit and compliance reporting. Using AWS CloudTrail and AWS Security Hub integration allows security teams to provide evidence of the effectiveness of controls for regulators or internal governance.

This approach supports the following Secure by Design principles:

- 5. Build in detect and respond security
- 9. Embed continuous assurance
- 10. Make changes securely

4. Use of third-party IDS and IPS solutions

When AWS-native services do not fully meet your security or functional requirements, you may need to use third-party solutions. Secure by Design requires that these are subject to thorough due diligence before they're adopted. Due diligence should include:

- assessment of supply chain risk
- vendor security posture
- update and patching mechanisms
- the product's suitability for the intended operational environment

This approach supports Secure by Design principle 2, Source secure technology products.

These solutions should integrate with AWS monitoring and logging frameworks, including [Amazon CloudWatch](#) and AWS Security Hub, to provide centralised visibility. Any deployment decisions should be risk-driven.

Selecting third-party tools also requires careful consideration of operational complexity and their day-to-day usability. Teams should assess how alerts are generated, prioritised and presented, including their format, severity models and whether they align with existing incident response processes.

Management consoles and administrative interfaces should be intuitive, role-aware and support secure access patterns, including integration with [AWS Identity and Access Management \(IAM\)](#) or federated identity providers. Configuration and tuning activities should be straightforward and well documented, so that teams can maintain and adjust controls without too much specialist effort or any risk of misconfiguration.

How well third-party solutions integrate existing AWS services is a key factor in determining whether they're suitable for your organisation. Third-party tools should:

- support native ingestion of logs, metrics and events from services such as CloudWatch, CloudTrail and VPC Flow Logs
- be able to surface findings into centralised platforms like AWS Security Hub

Strong integration reduces operational overhead, improves visibility across environments and helps ensure that security teams can respond effectively without introducing alert fatigue or fragmented workflows.

Finally, it's also essential to monitor third-party solutions on an ongoing basis. Regular vulnerability assessments, patch management and review of vendor security updates help maintain confidence in the effectiveness of IDS and IPS deployments.

This approach supports the following Secure by Design principles:

- 4. Design usable security controls

- 9. Embed continuous assurance
- 10. Make changes securely

5. IDS architecture and traffic visibility

Effective detection requires visibility into network and host activity. In AWS, VPC Traffic Mirroring, VPC Flow Logs and CloudTrail allow monitoring without impacting live workloads.

Host-based detection tools complement this by monitoring operating system (OS) behaviour and file integrity. Layered visibility supports defence in depth and means detection is effective even with encrypted or segmented traffic.

This approach supports the following Secure by Design principles:

- 5. Build in detect and respond security
- 6. Design flexible architectures
- 8. Defend in depth

Architectural design should make sure there's coverage across all traffic paths, including internal, east-west traffic between subnets. Using traffic mirroring or VPC ingress or egress inspection points reduces the risk that malicious activity could bypass detection.

Complementary monitoring of AWS-managed services such as [Amazon Relational Database Service \(RDS\)](#) or [AWS Lambda](#) means detection is not limited to EC2 instances. Secure by Design encourages comprehensive visibility across systems and services to reduce blind spots in modern cloud architectures.

This approach supports the following Secure by Design principles:

- 3. Adopt a risk-driven approach
- 5. Build in detect and respond security
- 8. Defend in depth

6. IPS architecture and inline enforcement

Intrusion prevention adds enforcement into traffic paths and so it should be designed to maintain availability as well as resilience.

In AWS, you should deploy IPS controls using highly available architectures across multiple Availability Zones (AZs) to avoid single points of failure. Services such as AWS Network Firewall and AWS WAF provide managed, scalable enforcement at network and application boundaries.

Rules should start in monitoring mode to observe traffic patterns and avoid accidental disruption. By moving to enforcement gradually, you can allow teams to test rules, fine-tune their thresholds and adjust responses.

Secure by Design emphasises proportionality and usability. This means that prevention controls should not degrade the performance of a service or its operational efficiency.

This approach supports the following Secure by Design principles:

- 3. Adopt a risk-driven approach
- 4. Design usable security controls
- 7. Minimise the attack surface

IPS architecture should also integrate with automated response and remediation tools.

For example, services like EventBridge or Lambda can:

- enforce quarantines
- deny traffic
- trigger alerts when threats are detected

This approach builds continuous detection and response into the service, and aligns with Secure by Design principles that promote resilience and active risk management.

This approach supports the following Secure by Design principles:

- 5. Build in detect and respond security
- 9. Embed continuous assurance

7. Logging, monitoring and evidence collection

Logging and monitoring form the foundation for detection and response. AWS recommends centralised collection of IDS and IPS logs using [Amazon CloudWatch Logs](#), AWS CloudTrail, and VPC Flow Logs. These logs provide tamper-evident records that support forensic analysis, continuous assurance and regulatory compliance.

Monitoring dashboards should provide near real-time visibility into threats, anomalies and system health. AWS Security Hub lets you aggregate findings from multiple accounts and services, which means you can provide operational teams with actionable insights.

Secure by Design stresses that controls should be usable to be effective, which in turn prevents alert fatigue and operational blind spots.

This approach supports the following Secure by Design principles:

- 4. Design usable security controls
- 5. Build in detect and respond security

You should collect evidence for both operational and audit purposes. Regularly validating logs, using retention policies and implementing access controls makes sure that historical activity is secure and available for investigation. When you combine automated logging with regular reviews, you're supporting resilience, accountability and ongoing risk management.

This approach supports the following Secure by Design principles:

- 1. Create responsibility for cyber security risk
- 3. Adopt a risk-driven approach
- 9. Embed continuous assurance

8. Alerting, response and automation

Detection is only valuable if it leads to a timely response. AWS services like Amazon EventBridge and AWS Security Hub allow centralised alerting and orchestration, so that IDS and IPS findings trigger defined workflows. Secure by Design emphasises integrating detection with operational processes so that teams can take action promptly.

This approach supports the following Secure by Design principles:

- 4. Design usable security controls
- 5. Build in detect and respond security

Automation improves consistency and speed of response, including actions such as:

- network isolation
- automated remediation
- notification escalation

For example, AWS Lambda functions can automatically apply firewall rule updates in response to validated threat intelligence. This makes you less reliant on manual intervention and helps make mitigation of threats proportional and controlled.

This approach supports the following Secure by Design principles:

- 6. Design flexible architectures
- 9. Embed continuous assurance
- 10. Make changes securely

By regularly testing alerting and automated response systems you can make sure they're reliable in realistic conditions. Carry out simulated attacks and red-team exercises to check responses are accurate, timely and do not introduce service degradation (as described in the Secure by Design activity [Discovering vulnerabilities](#)). This continuous validation aligns with Secure by Design guidance to embed active monitoring and risk-driven assurance.

9. Rule management and operational tuning

As workloads, threats and business priorities evolve, IDS and IPS rules require ongoing tuning to remain effective. AWS supports operational management through tools like [AWS Config](#) for configuration monitoring and compliance checks.

Secure by Design emphasises that controls must remain usable, proportionate and aligned with risk through the following principles:

- 4. Design usable security controls
- 9. Embed continuous assurance

You can reduce false positives and negatives with continuous reviews of detection and prevention rules. Use threat intelligence feeds and historical logs to inform updates. And by integrating with AWS-managed services, you can make sure rules are implemented consistently across accounts and regions.

This approach supports the following Secure by Design principles:

- 3. Adopt a risk-driven approach
- 5. Build in detect and respond security
- 9. Embed continuous assurance

To prioritise rules that matter most, it's essential that there's collaboration between security teams, operations and business units. Secure by Design encourages champions to engage with stakeholders so that controls are aligned with organisational priorities. For more on this, see the guidance on [establishing an internal Secure by Design working group](#).

This approach supports Secure by Design principle 4, Design usable security controls.

10. Resilience, scalability and change management

Since AWS workloads frequently scale, IDS and IPS systems must remain secure as infrastructure changes. By using multi-AZ deployments, auto-scaling and high availability architectures, you can make sure your organisation's systems stay resilient.

Secure by Design emphasises that systems must maintain security during operational change, failure or scaling events through the following principles:

- 6. Design flexible architectures
- 7. Minimise the attack surface
- 9. Embed continuous assurance

Infrastructure as Code (IAC), using [AWS CloudFormation](#) or [Terraform](#), enables repeatable deployments, version control and controlled updates to IDS and IPS components. This reduces configuration drift and helps you apply security controls consistently.

Your change management processes should include:

- testing
- validation
- approval of rule updates or architectural adjustments

Secure by Design encourages deliberate, auditable changes to prevent inadvertent weakening of security. By combining automated testing with formal approvals, you can ensure assurance is continuous across scaling operations.

11. Continuous assurance and validation

Continuous assurance validates that IDS and IPS controls are working effectively throughout the service life cycle. For evidence that security controls are operating as intended, you can use:

- dashboards
- metrics
- reviews
- AWS services like Security Hub, CloudWatch and GuardDuty, which support centralised monitoring and reporting

You can confirm that controls are effective under realistic conditions with regular testing activities, including penetration testing, red-team simulations and audit exercises. Secure by Design emphasises that monitoring and detection should be iterative, continuously improving with lessons learned and threat evolution.

Validation should also include integration testing between AWS-native and third-party solutions. By making sure performance is consistent and there's interoperability across accounts and regions, you can keep operations resilient and auditable.

This aligns with the Secure by Design approach of embedding assurance into every life cycle stage and supports the following Secure by Design principles:

- 3. Adopt a risk-driven approach
- 5. Build in detect and respond security
- 9. Embed continuous assurance

More information

- [Secure by Design activities \(UK Government Security\)](#)

- [Threat modelling \(NCSC\)](#)
- [Protecting internet-facing services on public service CNI \(NCSC\)](#)
- [Detecting the unknown - a guide to threat hunting \(Home Office Digital\)](#)
- [Logging and protective monitoring \(NCSC\)](#)